

### 7.3 ATOMIC STRUCTURE P2, P3&P5 QUESTIONS

Compiled by: R. Muza

Q1:JUN 2013 P2

7 (a) Define

(i) *binding energy of a nuclide,*

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(ii) *mass defect.*

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[4]

(b) An atom of an isotope of potassium,  $^{39}_{19}\text{K}$ , has a mass of 38.9533 u.

Given that the mass of a proton is 1.0073 u and that of a neutron is 1.0087 u, calculate

(i) the binding energy for the  $^{39}_{19}\text{K}$  isotope in MeV,

*binding energy* = \_\_\_\_\_

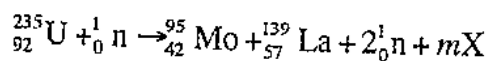
(ii) the binding energy per nucleon for the  $^{39}_{19}\text{K}$ .

*binding energy per nucleon* = \_\_\_\_\_

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- (c) Uranium-235 can split into molybdenum-95 and Lanthanum-139 by a process called nuclear fission. Two neutrons and unidentified particles are emitted in the process, as follows:



- (i) Define *nuclear fission*.

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- (ii) State how nuclear fission affects binding energy per nucleon.

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- (iii) Identify particle X and the integral value of  $m$ .

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[4]

#### Q2:JUN 2009 P5

- 5 (a) Explain the terms mass defect and binding energy. [2]

- (b) (i) Sketch a graph to show how the binding energy per nucleon varies with mass number.

- (ii) Explain the significance of the shape of the graph. [5]

- (c) Calculate the

- (i) energy released,

- (ii) the fraction of energy taken by the Beta particle.

when strontium-90 disintegrates by Beta emission to produce yttrium 90.

**Note:** mass of strontium nucleus is 89.90730 u  
mass of yttrium nucleus is 89.90670 u  
mass of Beta particle is 0.00055 u

[4]

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- (d) The mass of a cold body is measured on a sensitive balance. After heating the body strongly, its mass increased.

Explain this observation.

[1]

#### Q3:JUNE 2016 P5

- 2 (b) (i) Define *binding energy of a nucleus*.
- (ii) Sketch the graph of *binding energy per nucleon* against the *nucleon number* identifying the peak.
- (iii) Explain why uranium 238 would naturally undergo fission instead of fusion reactions.

[5]

#### Q4:JUN 2013 P5

- 2 (a) (i) State the difference between nuclear fusion and nuclear fission.
- (ii) Describe the significance of binding energy per nucleon to the processes of fission and fusion.
- (iii) Find the values of  $x$  and  $Y$  in the nuclear reaction below.
- $${}_{92}^{235}\text{U} + {}_0^1\text{n} \rightarrow {}_{56}^{143}\text{Ba} + {}_{36}^{90}\text{Kr} + Y{}_0^1\text{n} + Q$$
- (iv) Explain why the neutron produces the reaction in (iii).
- (v) The table shows some nuclide data.

| nuclide | mass / u |
|---------|----------|
| U-235   | 235.044  |
| Ba- 143 | 143.073  |
| Kr- 90  | 89.080   |
| n       | 1.009    |

Use the data to determine the energy  $Q$  in a(iii) in MeV.

[8]

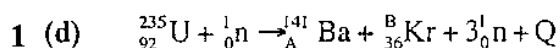
- (b) The reaction in a(iii) can be used in the generation of electricity.
- (i) Suggest how electricity can be generated from this reaction.
- (ii) Give **one** advantage and **one** disadvantage of generating electricity using this reaction.

[4]

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#### Q5:JUN 2011 P5



- (i) Find the values of A and B.
- (ii) Suggest why the neutron is capable of producing the fission above.
- (iii) Determine Q in MeV given the data below. [8]

| Element      | Atomic mass/u |
|--------------|---------------|
| Uranium -235 | 235.1         |
| Barium -141  | 141.0         |
| Krypton -93  | 91.9          |
| Neutron      | 1.009         |

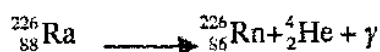
#### Q6:NOV 2003 P3

- 7 (a) (i) Describe qualitatively the alpha-scattering experiment. What evidence does it provide about the atom? [4]

- (ii) Calculate the distance of closest approach of an  $\alpha$ -particle of mass  $6.7 \times 10^{-27} \text{ kg}$  and velocity  $3.0 \times 10^7 \text{ ms}^{-1}$  when incident head-on with a stationary gold nucleus  $\left({}_{79}^{197}\text{Au}\right)$  (Assume the particles are in vacuum).

Explain your working. [6]

- (b) Radium (Ra) decays to radon (Rn) by the following nuclear reaction.



The atomic masses are:

|        |   |            |
|--------|---|------------|
| Radium | = | 226.0254 u |
| Radon  | = | 222.0175u  |
| Helium | = | 4.0026u    |

- (i) Calculate
  - 1 the energy, in MeV, released when a radium atom decays.
  - 2 the wavelength of the gamma ( $\gamma$ ) photon emitted during this decay given that 4% of the energy released is converted to  $\gamma$ -radiation.
- (ii) Account for the remaining 96% of the energy. [7]

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- (c) Calculate the speed of the  $\gamma$ -photons given that momentum is conserved during the decay. [3]

Q7:NOV 2002 P3

- 6 (e) The radioactive nuclei  $^{210}_{84}\text{Po}$  emit alpha particles of a single energy, the product nuclei being  $^{206}_{82}\text{Pb}$ .
- (i) Using the data below, calculate the energy in MeV, released in each decay.
- (ii) Explain why the alpha particle does not take up all this energy as its kinetic energy.
- (iii) Calculate the kinetic energy of the alpha particle taking the integer values of the nuclear masses.

| <u>Nucleus</u>         | <u>Mass(u)</u> |
|------------------------|----------------|
| $^{210}_{84}\text{Po}$ | 209.936 730    |
| $^{206}_{82}\text{Pb}$ | 205.929 421    |
| $\alpha$ - particle    | 4.001 504      |

1 unified atomic mass unit, u = 931 MeV

### 7.3 ATOMIC STRUCTURE P2, P3&P5 QUESTIONS

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**Q9:JUN 2008 P3**

- 7 (a) Distinguish between *radioactivity* and *nuclear fission*. [4]
- (b) A neutron is used to break a Uranium 235 atom into strontium and xenon atoms. The reaction can be represented as

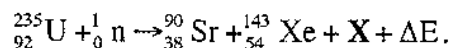


Table 7.1 contains same information concerning this reaction.

**Table 7.1**

| atom/particle | mass/u    |
|---------------|-----------|
| proton        | 1.00728   |
| neutron       | 1.00807   |
| helium -4     | 4.00150   |
| uranium -235  | 236.90957 |
| strontium     | 90.62019  |
| xenon         | 144.05770 |

- (i) Name the process represented by the equation.
- (ii) Identify particle X.
- (iii) Calculate the energy  $\Delta E$  released during the reaction.
- (iv) Explain why neutrons are preferred to protons as bombarding elementary particles in such reactions.
- (v) Suggest **one** useful application of such reactions. [9]
- (c) The energy released in (b)(iii) is sometimes called binding energy of a nucleus.
- (i) Define *binding energy*.
- (ii) Explain
- the link between binding energy per nucleon and the stability of a nucleus,

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2.      how binding energy per nucleon leads to nuclear fission  
         and nuclear fusion. [6]
- (d)    State **one** safety precaution which must be taken when handling  
         radioactive materials. [1]